Fuzzy Data Analysis: Theory and Applications in Natural Sciences

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Outline

1. Motto
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3. Need for fuzzy sets
4. Applications
5. Final word

Motto:

"The fuzzy set was conceived as a result of an attempt to come to grips with the problem of pattern recognition in the context of imprecisely defined categories. In such cases, the belonging of an object to a class is a matter of degree, as is the question of whether or not a group of objects form a cluster."

The founder of Fuzzy Logic, Prof. Lotfi A. Zadeh, University of Berkeley (1981)

Mathematics in Natural Sciences

"In any branch of study of the natural world, the amount of actual science contained therein is directly proportional to the amount of mathematics used. Chemistry can under no circumstances be regarded as a science."

Immanuel Kant (1724-1804)

Need for Fuzzy Sets – The bald man paradox

- $P(n)$ = "A person with n hairs on his head is bald"
- Prove by induction that all the people are bald
- Base case: $P(0)$ is obviously true
- Induction case: $P(n) \Rightarrow P(n+1)$
- If someone with n hairs on his head is bald, then by simply adding one hair, he will still be bald
- So, $\forall n \in \mathbb{N}\ P(n)$, i.e. everybody is bald (?!)
- The concept *bald* is a fuzzy concept

Applications of Fuzzy Data Analysis

- Fuzzy table of chemical elements
- Optimal selection of solvent systems
- Roman pottery (terra sigillata)
- Greek muds and pelloids
- Comparison of Romanian and American coals
- Quality of Danube water
- Characterization of Romanian medicinal herbs
- Mineral waters classification
- Characterization of medicinal plant extracts
- Intramolecular interactions and catalyst modeling
- Assesment of heart disease
- Intelligent disease identification
- Prognostic factors in liver failure in children
- Electric network distribution systems
- Undergraduate B.Sc. graduation examinations
- Software engineering projects of undergraduate students

Fuzzy table of chemical elements – the problem

- cluster structure of the chemical elements, based on their physical, chemical, and structural properties; results obtained are consistent with the chemical behavior and the electronic structure
- most properties determined by the outer electron configurations of their atoms; these configurations are the basis for the distribution of elements in the periodic system; is the classical structure of the periodic system really fully concordant with the physical and chemical properties?
- Mendeleev's classification was not founded on electron configuration but on observable data concerning physico-chemical properties of elements
- properties of elements are gradual by nature; they should be grouped in classes with no clearly-cut boundaries; each element should belong mainly to a single class and have smaller membership to all other classes

Fuzzy table of chemical elements – the methods

- fuzzy divisive hierarchical clustering based on 10 physical properties
  - based on 8 physical properties (excluding melting and boiling points)
  - based on 7 physical properties (excluding atomic mass)
  - based on 10 physical, chemical and structural properties
- fuzzy divisive hierarchical cross-clustering based on 10 physical properties
  - based on 10 physical, chemical and structural properties
- fuzzy divisive hierarchical characteristics clustering of the 10 physical properties
  - of the 10 physical, chemical and structural properties

Fuzzy table of chemical elements – conclusions

- on a chemical basis as well on a physical one the same partitions are almost invariably found; these essentially agree to the traditional chemical classifications, although the fuzzy membership details may differ
- the classification details as the membership degrees to various classes and the distances between elements point out some less trivial similarity relationships between the elements; these could suggest some new ideas with a view to possible applications in different fields
- at every classification level there are certain physical and chemical properties that influence the separation
- fuzzy characteristics clustering procedures revealed a high similarity between chemical and structural properties and also between the latter and some physical properties

B.Sc. graduation exam – the problem

- the relation between the quality of the programs and the style of the programmer
- it was considered that the style/knowledge of the programmer affects his efficiency, and the correctness of his programs
- data provided by the written B.Sc. graduation exam: final grade, grades for the programs, grades for general accepted rules for good programming (comments, indentation, names, design)
- use (Fuzzy) PCA to have a high quality 2D visual representation of the data
- use FDH clustering to get the clustering hierarchy, fuzzy membership degrees, and student-class relationships

B.Sc. graduation exam – the solution

- three homogeneous classes of good students; one homogeneous class of average students; a heterogenous class with worse prepared students
- over-average students group in homogeneous classes, i.e. they have a few, common patterns of approaching the exam subjects
- low graded students group in an heterogeneous class of isolated points, i.e. they have individual patterns of approaching the exams
- first principal component, associated to the final grade, covers 85% of the overall information in our data set
- the final grade better discriminates between the students than any other partial grade; this suggests that the problems have been properly chosen.

Software projects of undergraduate students – dependence of software attributes

- software projects produced by second year undergraduate students as part of their requirements curriculum and graded by graduate students as part of the “Software Metrics” course requirements
- factorial analysis, principal components analysis and fuzzy clustering
- discordance between the function points and size: not all projects were fully implemented
- most attributes depend on the general knowledge of the programmer; other factors: problem complexity and programmer discipline
- a good programming style and a correct programming habbit must be taught in parallel

Software projects of undergraduate students – undergraduate vs. graduate projects

- software projects produced by second year undergraduate students as part of their requirements curriculum
- software projects produced by graduate students working for their companies
- the undergraduate students of the second year learned to specify, design, code and test a complete program
- the master students learned to analyse software documents, to measure software attributes
- detect the proper relations among the data; the fuzzy cluster substructure of the data set and of the attributes set

Software projects of undergraduate students – undergraduate vs. graduate projects

- object-oriented attributes for procedural projects: (a) value of zero; (b) marked as missing; identical results
- industrial projects grouped separately, along patterns of similarity to student projects
- attributes generally split along lines of object-orientedness; apparently, the object-oriented attributes are consistent with the function points; the students follow the programming rules and habbits in the same manner, without regard to a procedural or OO project
- OO software metrics we used actually measure the complexity of the project; we need OO metrics to deal with the quality of an OO project

Software projects of undergraduate students – tracking mistakes in measurement

- software projects produced by second year undergraduate students as part of their requirements curriculum and graded by graduate students as part of the “Software Metrics” course requirements
- close dependence between the clarity, readability and correctness of a program
- strong correlation exists between comprehensibility and good design, confirmed by our experiment
- cluster analysis: izolated classes, with isolated points
- the general relation between certain metrics do not apply for those projects

Software projects of undergraduate students – tracking mistakes in measurement

- reason, incorrect evaluations of the attributes
- incorrect data typing: number of classes larger than number of lines
- number of comment lines equal to 2/3 of the number of lines
- small values for changeability, structuredness or comprehensibility, uncorrelated with large values for the other related attributes
- readability, comprehensibility, changeability, extensibility, maintainability are strongly dependent; this relation is not observed for the incorrectly graded projects

"[. . .] When I read (and reread) their papers and discuss them with my students, I am often struck by the whimsical image of Mendeleev sitting at a computer, entering his collected elemental properties into the fuzzy classification programs developed by the Cluj team, and assessing the various elemental relationships at different levels of fuzziness! How much sooner would eka-aluminum and other new elements have been isolated had Mendeleev had such tools at his fingertips? [. . .]"

Prof. Thomas Cundari, University of Memphis (2000)
in the foreword to

Horowitz, Sârbu, Pop, *Rational Classification of the Chemical Elements*
Dacia Publ. Co, Cluj-Napoca, 2000